Mingyue Xu

mx2258@columbia.edu | xu1864@purdue.edu
Google Scholar

Education

Purdue University West Lafayette, IN
Ph.D. Student in Computer Science *Aug 2023 – present*

- Advisor: Steve Hanneke

Columbia University New York, NY
M.S. in Data Science *Sep 2021 – Dec 2022*

- Advisor: Daniel Hsu
- GPA: 3.73/4.00

Shanghai Jiao Tong University Shanghai, China
B.S. in Mathematics *Sep 2017 – Jun 2021*

- GPA: 3.62/4.00

Research Interests

Statistical Learning Theory, Algorithmic Statistics, Deep Learning, Artificial Intelligence

Publications & Pre-prints/Under Review

(α-β) denotes alphabetical ordering, * denotes equal contribution.

S. Hanneke, H. Wang, **M. Xu** (α-β). "Optimal Learning under Tsybakov Noise". Submitted to the 67th IEEE Symposium on Foundations of Computer Science (FOCS 2026).

S. Hanneke and **M. Xu** (α-β). "When More Data Doesn't Help: Limits of Adaptation in Multitask Learning". The Fourty-third International Conference on Machine Learning (ICML 2026).

M. Xu, G. Vardi, I. Safran. "To Grok Grokking: Provable Grokking in Ridge Regression". The Fourty-third International Conference on Machine Learning (ICML 2026 – Oral presentation).

S. Hanneke and **M. Xu** (α-β). "Universal Rates of ERM for Agnostic Learning". The Thirty-eighth Annual Conference on Learning Theory (COLT 2025).

G. Yuan*, **M. Xu***, S. Kpotufe, D. Hsu. "Efficient Estimation of the Central Mean Subspace via Smoothed Gradient Outer Products". SIAM Journal on Mathematics of Data Science (SIMODS 2025).

S. Hanneke and **M. Xu** (α-β). "Universal Rates of Empirical Risk Minimization". The Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024).

J. Tu, W. Liu, X. Mao, **M. Xu**. "Distributed Semi-supervised Sparse Statistical Inference". IEEE Transactions on Information Theory (TIT 2023).

Work Experience

Machine Learning Research Intern Nov 2020 – Jun 2021
Qutronix Co., Ltd. *Shanghai, China*

Teaching Experience

CS182 Foundations of Computer Science — Spring 2025, Spring 2026
Teaching Assistant — *Purdue University*

CS251 Data Structures and Algorithms — Fall 2023, Spring 2024, Fall 2024
Teaching Assistant — *Purdue University*

CS381 Introduction to the Analysis of Algorithms — Fall 2025
Teaching Assistant — *Purdue University*